

SCHMUECKE, Germany

WMO: 105520

Lat: 50.6544N

Lon: 10.7694E

Elev: 3081

StdP: 13.13

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.9	8.9	-1.4	5.7	10.1	3.5	7.4	10.7	38.5	38.3	33.0	33.4	14.5	50	0.766

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	74.4	60.8	71.4	59.7	68.4	58.3	62.8	70.8	61.0	68.9	59.4	66.4	9.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.7	85.8	65.4	58.0	80.5	63.6	56.5	76.3	62.2	30.0	70.9	28.6	69.3	27.4	66.2	69.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.1	27.3	24.2	3.5	80.8	9.0	2.6	-3.1	82.7	-8.3	84.2	-13.4	85.7	-20.0	87.6
			3.0	66.9	8.9	1.7	-3.4	68.1	-8.6	69.1	-13.6	70.0	-20.0	71.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.8	25.8	24.6	31.7	37.7	47.8	52.0	56.8	56.4	50.1	43.0	33.3	29.0
	DBStd	13.52	8.42	8.88	7.69	7.67	7.45	7.47	7.02	6.56	6.60	6.95	7.36	6.56
	HDD50	4099	749	710	568	374	133	63	17	17	79	238	500	651
	HDD65	8857	1214	1130	1032	819	534	394	266	274	446	682	950	1116
	CDD50	741	0	0	1	5	64	124	227	216	83	21	0	0
	CDD65	26	0	0	0	0	0	5	12	9	0	0	0	0
	CDH74	102	0	0	0	0	1	15	53	33	0	0	0	0
	CDH80	6	0	0	0	0	0	0	4	2	0	0	0	0
	WSAvg	11.9	14.7	13.3	13.1	10.6	10.4	10.4	9.9	10.0	11.5	12.8	12.7	13.0
	PrecAvg	36.0	3.3	2.8	2.7	2.2	2.9	3.0	3.9	2.8	3.1	2.7	3.2	3.7
(15) Precipitation	PrecMax	46.5	7.1	7.2	5.5	5.2	8.9	5.2	6.9	7.4	6.7	8.5	5.8	9.3
	PrecMin	25.7	0.1	0.4	0.6	0.2	0.6	0.7	1.5	0.7	0.6	0.8	0.0	1.0
	PrecStd	5.3	1.6	1.5	1.5	1.1	1.8	1.3	1.2	1.3	1.4	1.6	1.4	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	47.3	56.6	61.1	69.8	76.3	79.6	79.3	71.5	64.8	53.9	46.6
		MCWB	39.7	42.3	45.9	48.5	55.9	62.6	63.3	62.9	58.5	54.5	44.3	42.8
	2%	DB	39.9	43.2	50.1	56.7	66.1	71.5	74.7	73.6	67.2	59.5	48.6	42.5
		MCWB	37.2	40.3	43.6	47.3	54.4	59.7	60.8	60.7	58.5	52.3	42.9	40.5
	5%	DB	37.3	39.5	45.5	53.1	62.8	67.5	71.6	70.3	63.4	55.5	46.3	40.3
		MCWB	36.4	37.1	41.2	45.2	52.8	57.6	60.2	59.3	56.9	50.5	42.3	38.7
	10%	DB	35.3	35.6	41.3	49.2	59.3	63.8	68.5	66.7	59.7	52.1	43.7	37.8
		MCWB	34.4	33.9	39.1	43.1	51.0	55.3	59.0	57.8	54.7	49.1	40.9	36.7
	0.4%	WB	41.6	43.8	46.9	50.7	58.0	63.9	65.6	64.1	62.2	56.6	47.6	44.4
		MCDB	42.2	45.9	52.2	57.3	66.1	71.4	75.4	75.6	66.2	62.0	49.4	44.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.3	40.8	44.3	48.0	55.5	60.9	63.2	61.9	59.6	53.8	45.2	41.6
		MCDB	38.6	42.5	49.4	55.9	62.7	69.0	71.4	70.7	64.9	57.0	46.9	42.0
	5%	WB	36.4	37.2	41.7	45.6	53.6	58.8	61.2	60.3	57.5	51.6	43.5	39.0
		MCDB	36.9	38.8	44.8	51.8	61.0	65.6	69.1	68.3	61.9	55.1	45.3	39.6
	10%	WB	34.5	34.0	39.3	43.8	51.9	56.6	59.5	58.7	55.7	49.8	41.1	36.9
		MCDB	35.2	35.5	41.3	48.8	58.8	62.5	66.6	65.3	59.2	51.8	43.4	37.5
	5% DB	MDBR	7.3	8.2	8.3	11.2	12.8	11.9	13.3	12.3	10.4	9.0	7.0	6.3
		MCDBR	9.0	11.7	13.7	16.8	17.5	16.7	17.5	17.3	15.1	13.8	12.0	9.3
		MCWBR	8.2	8.4	8.6	9.9	9.8	9.4	9.0	9.1	9.0	8.7	8.8	8.0
		MCWBR	8.6	10.4	12.2	15.7	15.7	14.9	16.0	15.0	13.3	11.1	10.4	8.0
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.2	8.3	8.6	10.0	9.5	9.1	9.2	8.7	8.8	7.8	8.4	7.6
(40) Clear-Sky Solar Irradiance	taub		0.278	0.298	0.332	0.365	0.369	0.367	0.379	0.373	0.344	0.322	0.301	0.275
	taud		2.451	2.418	2.379	2.322	2.330	2.345	2.309	2.350	2.422	2.484	2.484	2.448
	Ebn at Noon		245	268	277	278	281	282	276	272	269	255	233	228
	Edh at Noon		18	25	31	36	38	38	38	35	29	23	18	16
(44) All-Sky Solar Radiation	RadAvg		244	474	813	1288	1537	1676	1588	1369	984	575	270	184
	RadStd		43	71	107	166	181	141	192	140	138	71	38	22

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	+0.54	N/A	N/A	N/A	N/A

Nomenclature: See separate page